

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1239	0.0398	3.11 $\times$	0.1584	0.1785	0.89 $\times$
DBQuery	15	13	0.1188	0.0880	1.35 $\times$	0.1363	0.1649	0.83 $\times$
DecisionTree	7	6	3.632	2.925	1.24 $\times$	–	–	–
DomTree	15	14	0.1344	0.0610	2.21 $\times$	0.0576	0.0791	0.73 $\times$
KDTree	17	16	0.1745	0.1383	1.26 $\times$	–	–	–
LinearListReduction	11	11	0.0289	0.0049	5.84 $\times$	–	–	–
List	2	2	0.0884	0.0895	0.99 $\times$	0.2921	0.3607	0.81 $\times$
MonoTree	3	2	0.0172	0.0224	0.77 $\times$	0.0453	0.0428	1.06 $\times$
ObjectGraph	5	4	0.0878	0.0804	1.09 $\times$	0.1425	0.1611	0.88 $\times$
PiecewiseFunctions	8	7	0.1368	0.0905	1.51 $\times$	0.2396	0.2641	0.91 $\times$
TernaryTree	5	3	0.0278	0.0256	1.09 $\times$	0.0829	0.0800	1.04 $\times$
Trie	10	9	0.0631	0.0320	1.97 $\times$	0.1494	0.1665	0.90 $\times$

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.54 $\times$	0.52 $\times$	4.46 $\times$	0.77 $\times$	14.10 $\times$	0.47 $\times$	6.93 $\times$
DBQuery	1.05 $\times$	0.47 $\times$	1.28 $\times$	0.52 $\times$	8.31 $\times$	0.24 $\times$	6.26 $\times$
DecisionTree	1.24 $\times$	0.59 $\times$	16.91 $\times$	3.22 $\times$	0.90 $\times$	3.39 $\times$	37.30 $\times$
DomTree	1.50 $\times$	0.57 $\times$	3.16 $\times$	1.01 $\times$	21.04 $\times$	0.13 $\times$	10.72 $\times$
KDTree	1.26 $\times$	0.77 $\times$	1.26 $\times$	0.78 $\times$	0.23 $\times$	0.79 $\times$	4.96 $\times$
LinearListReduction	5.87 $\times$	0.90 $\times$	33.22 $\times$	5.35 $\times$	39.43 $\times$	5.25 $\times$	11.64 $\times$
List	0.82 $\times$	0.80 $\times$	3.55 $\times$	0.51 $\times$	0.10 $\times$	0.68 $\times$	1.46 $\times$
MonoTree	0.97 $\times$	0.85 $\times$	0.52 $\times$	0.76 $\times$	0.29 $\times$	0.65 $\times$	1.23 $\times$
ObjectGraph	0.94 $\times$	0.75 $\times$	5.03 $\times$	0.81 $\times$	0.53 $\times$	0.63 $\times$	2.86 $\times$
PiecewiseFunctions	1.10 $\times$	0.60 $\times$	2.85 $\times$	0.94 $\times$	2.23 $\times$	0.47 $\times$	4.48 $\times$
TernaryTree	1.13 $\times$	0.79 $\times$	0.32 $\times$	0.36 $\times$	0.20 $\times$	2.66 $\times$	1.38 $\times$
Trie	1.13 $\times$	0.56 $\times$	3.42 $\times$	0.87 $\times$	3.18 $\times$	0.12 $\times$	6.27 $\times$
Geomean	1.30 $\times$	0.67 $\times$	2.91 $\times$	0.95 $\times$	1.66 $\times$	0.67 $\times$	4.92 $\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.3e+08/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 2.9e+06	2.2e+05 /2.9e+05	1.2e+06/ 8.7e+04	1.3e+04 /2.8e+04	5.1e+07/ 7.3e+06
DBQuery	9.3e+08/ 8.9e+08	1.1e+09 /2.2e+09	3.9e+06/ 3.0e+06	1.7e+05 /3.3e+05	8.0e+05/ 9.6e+04	6.8e+03 /2.9e+04	2.1e+07/ 3.3e+06
DecisionTree	1.8e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.3e+08/ 3.2e+07	1.1e+05/ 3.5e+04	1.9e+06 /2.1e+06	1.1e+05/ 3.4e+04	1.1e+09/ 2.9e+07
DomTree	8.1e+08/ 5.4e+08	1.2e+09 /2.2e+09	1.2e+07/ 3.8e+06	1.1e+05/ 1.1e+05	1.1e+06/ 5.1e+04	4.3e+03 /3.3e+04	5.7e+07/ 5.3e+06
KDTree	8.8e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.5e+07/ 1.2e+07	7.7e+02 /9.8e+02	2.1e+04 /9.5e+04	7.8e+02 /9.8e+02	4.4e+07/ 8.9e+06
LinearListReduction	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	2.3e+02/ 4.3e+01	2.7e+05/ 6.9e+03	2.3e+02/ 4.4e+01	1.2e+07/ 1.1e+06
List	1.2e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.6e+06	3.0e+05 /5.9e+05	1.9e+04 /2.0e+05	9.6e+03 /1.4e+04	4.8e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.4e+05 /4.6e+05	3.5e+04 /4.6e+04	3.7e+03 /1.3e+04	1.5e+03 /2.3e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.2e+08 /8.8e+08	3.0e+09 /4.0e+09	9.0e+06/ 1.8e+06	2.1e+05 /2.6e+05	3.0e+04 /5.6e+04	1.1e+04 /1.8e+04	2.4e+07/ 8.3e+06
PiecewiseFunctions	1.2e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.5e+07/ 5.2e+06	4.5e+05 /4.8e+05	2.0e+05/ 8.8e+04	2.2e+04 /4.6e+04	4.4e+07/ 9.9e+06
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.1e+05 /1.6e+06	6.3e+04 /1.7e+05	4.2e+03 /2.1e+04	2.6e+03/ 9.6e+02	3.3e+06/ 2.4e+06
Trie	6.5e+08/ 5.7e+08	1.2e+09 /2.2e+09	9.2e+06/ 2.7e+06	2.0e+05 /2.3e+05	1.6e+05/ 5.2e+04	6.0e+03 /5.0e+04	2.5e+07/ 4.1e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2823	1.111	0.2183	<i>OOM</i>
DBQuery	0.2551	0.3269	0.2529	1.123
DecisionTree	3.632	5.621	2.925	25.22
DomTree	0.1921	0.3329	0.1400	0.5272
KDTree	0.1745	0.2435	0.1383	0.7226
LinearListReduction	0.0289	0.1391	0.0049	<i>OOM</i>
List	0.3805	<i>OOM</i>	0.4503	<i>OOM</i>
MonoTree	0.0626	0.1407	0.0652	0.2735
ObjectGraph	0.2303	0.4812	0.2415	0.7037
OctTree	0.5774	0.6404	–	<i>OOM</i>
PiecewiseFunctions	0.3764	0.5558	0.3546	2.161
TernaryTree	0.1108	0.1436	0.1056	0.6178
Trie	0.2125	0.2939	0.1985	1.050

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.29 $\times$	3.93 $\times$	5.09 $\times$	–
DBQuery	1.01 $\times$	1.28 $\times$	1.29 $\times$	0.29 $\times$
DecisionTree	1.24 $\times$	1.55 $\times$	1.92 $\times$	0.22 $\times$
DomTree	1.37 $\times$	1.73 $\times$	2.38 $\times$	0.63 $\times$
KDTree	1.26 $\times$	1.40 $\times$	1.76 $\times$	0.34 $\times$
LinearListReduction	5.84 $\times$	4.82 $\times$	28.11 $\times$	–
List	0.85 $\times$	–	–	–
MonoTree	0.96 $\times$	2.25 $\times$	2.16 $\times$	0.51 $\times$
ObjectGraph	0.95 $\times$	2.09 $\times$	1.99 $\times$	0.68 $\times$
OctTree	–	1.11 $\times$	–	–
PiecewiseFunctions	1.06 $\times$	1.48 $\times$	1.57 $\times$	0.26 $\times$
TernaryTree	1.05 $\times$	1.30 $\times$	1.36 $\times$	0.23 $\times$
Trie	1.07 $\times$	1.38 $\times$	1.48 $\times$	0.28 $\times$

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0118 $\pm$ 0.00001	0.0801 $\pm$ 0.00109	0.0019$\pm$0.00000	<i>OOM</i>	6.14 $\times$	6.80 $\times$	41.77 $\times$	–
branchStatsPass	F	2/9	78%	0.0114 $\pm$ 0.00001	0.0806 $\pm$ 0.00002	0.0036$\pm$0.00000	<i>OOM</i>	3.20 $\times$	7.05 $\times$	22.61 $\times$	–
castInstCountPass	F	3/9	67%	0.0179 $\pm$ 0.00089	0.0806 $\pm$ 0.00003	0.0013$\pm$0.00000	<i>OOM</i>	13.88 $\times$	4.50 $\times$	62.53 $\times$	–
has cycle	F	4/9	56%	0.0239 $\pm$ 0.00079	0.0951 $\pm$ 0.00002	0.0180$\pm$0.00014	<i>OOM</i>	1.33 $\times$	3.98 $\times$	5.29 $\times$	–
instCountPass	F	2/9	78%	0.0118 $\pm$ 0.00002	0.0812 $\pm$ 0.00002	0.0019$\pm$0.00000	<i>OOM</i>	6.06 $\times$	6.88 $\times$	41.68 $\times$	–
latencyModelPass	F	3/9	67%	0.0116 $\pm$ 0.00000	0.0813 $\pm$ 0.00002	0.0030$\pm$0.00000	<i>OOM</i>	3.93 $\times$	6.98 $\times$	27.47 $\times$	–
memoryOpStatsPass	F	3/9	67%	0.0115 $\pm$ 0.00001	0.0806 $\pm$ 0.00131	0.0038$\pm$0.00001	<i>OOM</i>	3.03 $\times$	6.98 $\times$	21.18 $\times$	–
stripSideEffectsPass	M	7/9	22%	0.0793$\pm$0.00142	0.1910 $\pm$ 0.00020	0.0885 $\pm$ 0.00164	<i>OOM</i>	0.90 $\times$	2.41 $\times$	2.16 $\times$	–
targetReturnPass	M	9/9	0%	0.0791$\pm$0.00017	0.1891 $\pm$ 0.00023	0.0900 $\pm$ 0.00011	<i>OOM</i>	0.88 $\times$	2.39 $\times$	2.10 $\times$	–
throughputModelPass	F	3/9	67%	0.0117 $\pm$ 0.00006	0.0806 $\pm$ 0.00086	0.0029$\pm$0.00000	<i>OOM</i>	3.96 $\times$	6.90 $\times$	27.32 $\times$	–
verifyIR	F	9/9	0%	0.0122 $\pm$ 0.00017	0.0706 $\pm$ 0.00154	0.0034$\pm$0.00024	<i>OOM</i>	3.56 $\times$	5.79 $\times$	20.59 $\times$	–
Total				0.2823	1.111	0.2183	–	1.29 $\times$	3.93 $\times$	5.09 $\times$	–
Geomean								3.14 $\times$	5.14 $\times$	16.14 $\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.0e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.2e+06/ 6.1e+03	1.3e+02/ 2.3e+01	2.2e+05/ 1.1e+03	1.3e+02/ 2.3e+01	4.7e+06/ 1.1e+03
branchStatsPass	F	5.8e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.2e+05/ 2.2e+04	2.4e+02/ 3.8e+01	4.9e+04/ 3.2e+03	2.4e+02/ 3.9e+01	4.7e+06/ 4.3e+05
castInstCountPass	F	9.0e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /2.2e+04	8.2e+01/ 2.1e+01	7.9e+02 /1.1e+03	8.1e+01/ 2.1e+01	4.6e+06/ 3.8e+03
has cycle	F	1.2e+08/ 9.1e+07	1.1e+08 /1.5e+08	2.3e+06/ 2.2e+05	3.5e+02/ 1.1e+02	1.5e+05/ 2.2e+04	3.6e+02/ 1.1e+02	5.1e+06/ 2.3e+06
instCountPass	F	6.0e+07/ 9.9e+06	5.6e+07 /5.7e+07	1.2e+06/ 1.6e+04	1.1e+02/ 2.7e+01	2.3e+05/ 1.1e+03	1.2e+02/ 2.7e+01	4.7e+06/ 1.3e+03
latencyModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.7e+04	2.9e+02/ 3.7e+01	2.0e+05/ 3.9e+03	3.0e+02/ 3.7e+01	4.7e+06/ 5.3e+05
memoryOpStatsPass	F	5.9e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.7e+05/ 3.1e+04	1.1e+02/ 3.9e+01	3.0e+04/ 3.2e+03	1.1e+02/ 4.2e+01	4.6e+06/ 4.4e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.8e+06/ 9.9e+05	1.1e+05 /1.5e+05	6.0e+03 /2.1e+04	4.2e+03 /7.6e+03	4.1e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.0e+08	1.7e+08 /6.3e+08	1.5e+06/ 1.5e+06	1.1e+05 /1.3e+05	5.5e+03 /2.3e+04	6.8e+03 /2.0e+04	4.2e+06/ 1.4e+06
throughputModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	2.6e+02/ 3.7e+01	2.0e+05/ 3.6e+03	2.6e+02/ 3.8e+01	4.7e+06/ 5.4e+05
verifyIR	F	6.2e+07/ 1.7e+07	7.4e+07 /8.6e+07	8.9e+05/ 1.9e+04	4.0e+02/ 1.0e+02	1.4e+05/ 4.1e+03	4.0e+02/ 1.0e+02	4.7e+06/ 4.5e+05
Total		9.3e+08/6.0e+08	9.9e+08/1.9e+09	1.3e+07/2.9e+06	2.2e+05/2.9e+05	1.2e+06/8.7e+04	1.3e+04/2.8e+04	5.1e+07/7.3e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/–	1.6e+08/–	2.2e+06/–	7.2e+02/–	9.1e+04/–	7.2e+02/–	1.4e+07/–
branchStatsPass	F	4.1e+08/–	1.8e+08/–	2.1e+06/–	6.8e+02/–	4.4e+04/–	6.8e+02/–	1.5e+07/–
castInstCountPass	F	4.1e+08/–	1.8e+08/–	2.1e+06/–	5.7e+02/–	4.5e+04/–	5.7e+02/–	1.5e+07/–
has cycle	F	4.8e+08/–	2.1e+08/–	2.5e+06/–	2.3e+02/–	7.0e+04/–	2.4e+02/–	1.7e+07/–
instCountPass	F	4.1e+08/–	1.6e+08/–	2.2e+06/–	3.8e+02/–	9.7e+04/–	3.8e+02/–	1.4e+07/–
latencyModelPass	F	4.1e+08/–	1.7e+08/–	2.1e+06/–	3.4e+02/–	4.5e+04/–	3.4e+02/–	1.4e+07/–
memoryOpStatsPass	F	4.1e+08/–	1.9e+08/–	2.0e+06/–	5.3e+02/–	4.2e+04/–	5.4e+02/–	1.5e+07/–
stripSideEffectsPass	M	7.1e+08/–	3.3e+08/–	2.0e+06/–	1.2e+05/–	2.2e+05/–	2.9e+04/–	2.4e+07/–
targetReturnPass	M	7.0e+08/–	3.2e+08/–	2.1e+06/–	1.0e+05/–	2.2e+05/–	3.5e+04/–	2.4e+07/–
throughputModelPass	F	4.1e+08/–	1.7e+08/–	2.1e+06/–	2.6e+02/–	4.4e+04/–	2.7e+02/–	1.5e+07/–
verifyIR	F	3.6e+08/–	1.2e+08/–	2.5e+06/–	4.3e+02/–	2.0e+05/–	4.4e+02/–	1.1e+07/–
Total		5.1e+09/–	2.2e+09/–	2.4e+07/–	2.3e+05/–	1.1e+06/–	6.9e+04/–	1.8e+08/–

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0684$\pm$0.00127	0.0759 $\pm$ 0.00010	0.0821 $\pm$ 0.00100	0.2974 $\pm$ 0.00315	0.83 $\times$	1.11 $\times$	0.92 $\times$	0.26 $\times$
countJoins	F	3/15	80%	0.0186 $\pm$ 0.00001	0.0285 $\pm$ 0.00001	0.0112$\pm$0.00000	0.0860 $\pm$ 0.00017	1.65 $\times$	1.53 $\times$	2.53 $\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0186 $\pm$ 0.00001	0.0282 $\pm$ 0.00001	0.0122$\pm$0.00001	0.0883 $\pm$ 0.00173	1.53 $\times$	1.51 $\times$	2.31 $\times$	0.32 $\times$
hashJoinPressure	F	5/15	67%	0.0190 $\pm$ 0.00079	0.0324 $\pm$ 0.00001	0.0124$\pm$0.00002	0.0870 $\pm$ 0.00255	1.52 $\times$	1.71 $\times$	2.60 $\times$	0.37 $\times$
scaleCosts	M	15/15	0%	0.0678$\pm$0.00011	0.0754 $\pm$ 0.00109	0.0828 $\pm$ 0.00009	0.2979 $\pm$ 0.00189	0.82 $\times$	1.11 $\times$	0.91 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0185 $\pm$ 0.00011	0.0282 $\pm$ 0.00011	0.0149$\pm$0.00007	0.0892 $\pm$ 0.00002	1.24 $\times$	1.52 $\times$	1.89 $\times$	0.32 $\times$
sumMemory	F	6/15	60%	0.0184 $\pm$ 0.00106	0.0284 $\pm$ 0.00001	0.0152$\pm$0.00074	0.0894 $\pm$ 0.00128	1.21 $\times$	1.54 $\times$	1.87 $\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0257 $\pm$ 0.00003	0.0298 $\pm$ 0.00002	0.0219$\pm$0.00001	0.0883 $\pm$ 0.00001	1.17 $\times$	1.16 $\times$	1.36 $\times$	0.34 $\times$
Total				0.2551	0.3269	0.2529	1.123	1.01 $\times$	1.28 $\times$	1.29 $\times$	0.29 $\times$
Geomean								1.21 $\times$	1.38 $\times$	1.67 $\times$	0.31 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.9e+05 /1.2e+06	9.8e+04 /1.6e+05	3.2e+04/ 2.3e+04	2.7e+03 /1.1e+04	2.5e+06/ 9.2e+05
countJoins	F	9.4e+07/ 5.7e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.6e+04	4.2e+02/ 9.1e+01	1.4e+05/ 2.1e+03	4.3e+02/ 5.4e+01	2.6e+06/ 2.5e+03
filterSelectivitySkew	F	9.4e+07/ 6.2e+07	1.0e+08 /1.1e+08	5.2e+05/ 7.2e+04	2.7e+02/ 9.7e+01	1.2e+05/ 6.3e+03	2.8e+02/ 9.7e+01	2.6e+06/ 2.2e+05
hashJoinPressure	F	9.6e+07/ 6.3e+07	8.4e+07 /9.9e+07	7.3e+05/ 1.0e+05	4.3e+02/ 1.1e+02	1.9e+05/ 7.5e+03	4.3e+02/ 1.1e+02	2.7e+06/ 1.1e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.6e+05 /1.3e+06	7.0e+04 /1.6e+05	3.1e+04/ 2.8e+04	2.3e+03 /1.7e+04	2.4e+06/ 1.1e+06
sumCost	F	9.4e+07/ 7.5e+07	9.0e+07 /1.2e+08	5.5e+05/ 8.1e+04	1.0e+02/ 6.1e+01	1.2e+05/ 1.1e+04	1.1e+02/ 6.7e+01	2.7e+06/ 3.2e+05
sumMemory	F	9.3e+07/ 7.7e+07	9.0e+07 /1.2e+08	6.7e+05/ 8.1e+04	3.4e+02/ 6.0e+01	1.6e+05/ 1.1e+04	3.4e+02/ 6.0e+01	2.7e+06/ 3.2e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.7e+05/ 1.1e+05	1.5e+02/ 1.1e+02	2.1e+04/ 7.7e+03	1.6e+02/ 1.1e+02	2.4e+06/ 2.8e+05
Total		9.3e+08/8.9e+08	1.1e+09/2.2e+09	3.9e+06/3.0e+06	1.7e+05/3.3e+05	8.0e+05/9.6e+04	6.8e+03/2.9e+04	2.1e+07/3.3e+06

Table 11: Per-pass PAPI counters for DBQuery (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	5.9e+05 /2.0e+07	8.9e+04 /6.8e+05	1.9e+04 /4.4e+04	2.5e+03 /5.4e+04	2.2e+06/ 9.8e+05
countJoins	F	1.4e+08 /4.4e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.1e+05	2.8e+02 /7.6e+02	1.1e+05/ 1.3e+04	2.8e+02/ 1.1e+02	2.3e+06/ 1.9e+04
filterSelectivitySkew	F	1.4e+08 /4.5e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.2e+05	2.0e+02 /1.0e+03	6.4e+04/ 1.7e+04	2.0e+02 /2.9e+02	2.3e+06/ 1.9e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.4e+05	2.0e+02 /7.7e+02	1.1e+05/ 1.6e+04	2.0e+02/ 1.9e+02	2.2e+06/ 1.0e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.2e+05 /2.3e+07	7.8e+04 /3.0e+05	1.6e+04 /5.4e+04	3.7e+03 /5.9e+04	2.3e+06/ 1.2e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.7e+06/ 1.2e+06	1.0e+02 /8.9e+02	7.4e+04/ 2.2e+04	1.0e+02 /4.9e+02	2.3e+06/ 3.2e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	3.5e+02 /9.7e+02	6.9e+04/ 2.1e+04	3.6e+02 /4.6e+02	2.3e+06/ 3.1e+05
sumRows	F	1.5e+08 /4.5e+08	2.3e+08 /8.3e+08	1.6e+06/ 1.2e+06	1.4e+02 /7.6e+02	5.8e+04/ 2.2e+04	1.4e+02 /4.7e+02	2.2e+06/ 3.2e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.8e+07	1.7e+05/9.8e+05	5.2e+05/2.1e+05	7.5e+03/1.1e+05	1.8e+07/3.4e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.114 $\pm$ 0.0044	4.258 $\pm$ 0.0044	2.583$\pm$0.0036	21.25 $\pm$ 0.010	1.21$\times$	1.37$\times$	1.65$\times$	0.20 $\times$
classify Depth	F	4/7	43%	0.0447 $\pm$ 0.00001	0.0609 $\pm$ 0.00001	0.0365$\pm$0.00006	0.2991 $\pm$ 0.00143	1.23$\times$	1.36$\times$	1.67$\times$	0.20 $\times$
classify tree	F	5/7	29%	0.0446 $\pm$ 0.00001	0.0609 $\pm$ 0.00096	0.0365$\pm$0.00090	0.3000 $\pm$ 0.00164	1.22$\times$	1.36$\times$	1.67$\times$	0.20 $\times$
countFeatureUses	F	3/7	57%	0.0477 $\pm$ 0.00001	0.1412 $\pm$ 0.00141	0.0318$\pm$0.00002	0.3783 $\pm$ 0.00116	1.50$\times$	2.96$\times$	4.44$\times$	0.37 $\times$
countLeaves	F	2/7	71%	0.0455 $\pm$ 0.00001	0.1393 $\pm$ 0.00136	0.0249$\pm$0.00001	0.3689 $\pm$ 0.00130	1.83$\times$	3.06$\times$	5.59$\times$	0.38 $\times$
countNodes	F	2/7	71%	0.0451 $\pm$ 0.00001	0.1395 $\pm$ 0.00146	0.0260$\pm$0.00003	0.3686 $\pm$ 0.00147	1.73$\times$	3.09$\times$	5.36$\times$	0.38 $\times$
countSmallLeaves	F	3/7	57%	0.0485 $\pm$ 0.00003	0.1353 $\pm$ 0.00002	0.0321$\pm$0.00001	0.3714 $\pm$ 0.00181	1.51$\times$	2.79$\times$	4.22$\times$	0.36 $\times$
inferenceCost	F	2/7	71%	0.0489 $\pm$ 0.00084	0.1379 $\pm$ 0.00141	0.0290$\pm$0.00001	0.3700 $\pm$ 0.00209	1.69$\times$	2.82$\times$	4.76$\times$	0.37 $\times$
sumImpurity	F	3/7	57%	0.0477 $\pm$ 0.00002	0.1381 $\pm$ 0.00006	0.0320$\pm$0.00002	0.3808 $\pm$ 0.00174	1.49$\times$	2.89$\times$	4.31$\times$	0.36 $\times$
sumPathLengths	F	3/7	57%	0.0504 $\pm$ 0.00002	0.1379 $\pm$ 0.00002	0.0339$\pm$0.00001	0.3872 $\pm$ 0.00112	1.49$\times$	2.73$\times$	4.07$\times$	0.36 $\times$
sumSamples	F	3/7	57%	0.0458 $\pm$ 0.00002	0.1340 $\pm$ 0.00130	0.0316$\pm$0.00002	0.3837 $\pm$ 0.00119	1.45$\times$	2.92$\times$	4.23$\times$	0.35 $\times$
treeDepth	F	2/7	71%	0.0487 $\pm$ 0.00072	0.1379 $\pm$ 0.00019	0.0280$\pm$0.00081	0.3699 $\pm$ 0.00174	1.74$\times$	2.83$\times$	4.93$\times$	0.37 $\times$
Total				3.632	5.621	2.925	25.22	1.24$\times$	1.55$\times$	1.92$\times$	0.22 $\times$
Geomean								1.49$\times$	2.40$\times$	3.58$\times$	0.32 $\times$

Table 13: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.6e+07	1.0e+05/ 2.9e+04	1.5e+06 /2.0e+06	1.0e+05/ 2.7e+04	9.2e+08/ 1.7e+07
classify Depth	F	2.3e+08/ 1.8e+08	4.7e+08 /8.5e+08	6.5e+06/ 3.7e+05	1.2e+03/ 5.8e+02	1.8e+04 /2.8e+04	1.2e+03/ 5.4e+02	1.3e+07/ 2.7e+05
classify tree	F	2.3e+08/ 1.9e+08	4.7e+08 /8.5e+08	6.5e+06/ 3.7e+05	9.5e+02/ 5.3e+02	2.2e+04 /2.8e+04	9.5e+02/ 5.0e+02	1.3e+07/ 2.8e+05
countFeatureUses	F	2.4e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.3e+06/ 1.1e+06	6.6e+02 /8.2e+02	3.6e+04/ 1.4e+04	6.6e+02 /8.2e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.2e+06/ 1.1e+05	1.0e+03/ 3.7e+02	4.4e+04/ 8.1e+03	1.0e+03/ 3.7e+02	1.4e+07/ 3.3e+05
countNodes	F	2.3e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.2e+06/ 1.1e+05	7.1e+02/ 3.1e+02	4.8e+04/ 8.1e+03	7.2e+02/ 3.1e+02	1.4e+07/ 3.5e+05
countSmallLeaves	F	2.5e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.7e+06/ 2.0e+05	7.5e+02 /7.9e+02	2.7e+04/ 1.3e+04	7.5e+02 /7.9e+02	1.6e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.2e+05	1.3e+03/ 4.0e+02	1.3e+05/ 7.9e+03	1.3e+03/ 4.0e+02	1.4e+07/ 3.3e+05
sumImpurity	F	2.4e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.5e+06/ 1.0e+06	1.0e+03/ 7.3e+02	3.5e+04/ 1.3e+04	1.0e+03/ 7.3e+02	1.5e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.4e+06/ 1.2e+06	1.0e+03/ 9.5e+02	3.6e+04/ 1.4e+04	1.0e+03/ 9.5e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.3e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.1e+06/ 4.6e+05	6.9e+02/ 6.8e+02	3.7e+04/ 1.4e+04	6.9e+02/ 6.7e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.1e+05	1.1e+03/ 4.8e+02	1.5e+04/ 8.1e+03	1.1e+03/ 4.7e+02	1.4e+07/ 3.3e+05
Total		1.8e+10/1.5e+10	4.0e+10/6.8e+10	5.3e+08/3.2e+07	1.1e+05/3.5e+04	1.9e+06/2.1e+06	1.1e+05/3.4e+04	1.1e+09/2.9e+07

Table 14: Per-pass PAPI counters for DecisionTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.4e+07 /4.5e+07	2.4e+04 /1.5e+05	3.6e+05 /2.1e+06	2.4e+04 /3.1e+04	8.5e+08/ 1.8e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.8e+05	3.0e+02 /1.5e+03	4.7e+03 /3.2e+04	3.0e+02/ 2.3e+02	1.2e+07/ 2.7e+05
classify tree	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.1e+05	3.0e+02 /1.6e+03	4.7e+03 /3.0e+04	3.0e+02/ 2.0e+02	1.2e+07/ 2.9e+05
countFeatureUses	F	7.1e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	8.9e+02 /2.1e+03	7.3e+03 /6.9e+04	8.9e+02/ 7.2e+02	9.3e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	6.6e+02 /2.6e+03	1.1e+04 /3.2e+04	6.6e+02/ 2.1e+02	9.2e+06/ 3.0e+05
countNodes	F	7.1e+08 /1.9e+09	1.2e+09 /3.0e+09	4.4e+06/ 6.8e+05	7.0e+02 /2.1e+03	8.1e+03 /3.2e+04	6.9e+02/ 4.4e+02	9.2e+06/ 3.1e+05
countSmallLeaves	F	6.8e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	1.1e+03 /1.9e+03	6.0e+03 /7.3e+04	1.1e+03/ 6.5e+02	9.3e+06/ 1.2e+06
inferenceCost	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.1e+05	6.9e+02 /1.4e+03	7.8e+03 /3.0e+04	6.8e+02/ 2.5e+02	9.2e+06/ 3.3e+05
sumImpurity	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	3.1e+02/ 2.6e+02	7.7e+03 /7.5e+04	3.2e+02/ 2.5e+02	9.4e+06/ 1.2e+06
sumPathLengths	F	7.0e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	4.0e+02 /3.0e+03	5.9e+03 /7.8e+04	4.0e+02 /8.1e+02	9.2e+06/ 1.1e+06
sumSamples	F	6.8e+08 /1.9e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	1.3e+03 /2.4e+03	5.3e+03 /7.7e+04	1.3e+03/ 8.9e+02	9.3e+06/ 1.2e+06
treeDepth	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	3.3e+02 /1.5e+03	7.8e+03 /3.1e+04	3.3e+02/ 2.2e+02	9.2e+06/ 3.3e+05
Total		2.8e+10/1.3e+11	6.4e+10/2.2e+11	4.0e+07/6.6e+07	3.1e+04/1.7e+05	4.4e+05/2.7e+06	3.1e+04/3.6e+04	9.6e+08/2.6e+07

Table 15: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0345 $\pm$ 0.00002	0.0683 $\pm$ 0.00007	0.0190$\pm$0.00003	0.0901 $\pm$ 0.00003	1.82$\times$	1.98$\times$	3.60$\times$	0.76 $\times$
computeWidths	M	14/15	7%	0.0294$\pm$0.00005	0.0300 $\pm$ 0.00003	0.0446 $\pm$ 0.00016	0.0867 $\pm$ 0.00006	0.66 $\times$	1.02 $\times$	0.67 $\times$	0.35 $\times$
count styled	F	3/15	80%	0.0334 $\pm$ 0.00001	0.0681 $\pm$ 0.00112	0.0116$\pm$0.00001	0.0880 $\pm$ 0.00002	2.88$\times$	2.04$\times$	5.87$\times$	0.77 $\times$
find max Bottom	F	5/15	67%	0.0333 $\pm$ 0.00001	0.0678 $\pm$ 0.00001	0.0188$\pm$0.00004	0.0878 $\pm$ 0.00132	1.78$\times$	2.03$\times$	3.61$\times$	0.77 $\times$
scaleLayout	M	15/15	0%	0.0282$\pm$0.00003	0.0301 $\pm$ 0.00005	0.0345 $\pm$ 0.00004	0.0862 $\pm$ 0.00007	0.82 $\times$	1.07 $\times$	0.87 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0331 $\pm$ 0.00002	0.0687 $\pm$ 0.00102	0.0116$\pm$0.00002	0.0883 $\pm$ 0.00115	2.85$\times$	2.07$\times$	5.92$\times$	0.78 $\times$
Total				0.1921	0.3329	0.1400	0.5272	1.37$\times$	1.73$\times$	2.38$\times$	0.63 $\times$
Geomean								1.56$\times$	1.63$\times$	2.53$\times$	0.59 $\times$

Table 16: Per-pass PAPI counters for DomTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.7e+08/ 9.6e+07	2.9e+08 /4.9e+08	2.7e+06/ 4.7e+05	3.3e+02/ 1.2e+02	1.7e+05/ 1.2e+04	3.3e+02/ 1.2e+02	1.4e+07/ 1.8e+06
computeWidths	M	6.5e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	5.1e+04 /6.3e+04	5.3e+03 /1.1e+04	1.3e+03 /1.8e+04	1.2e+06/ 4.3e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 2.4e+04	5.1e+02/ 7.3e+01	3.0e+05/ 5.0e+03	5.1e+02/ 7.6e+01	1.4e+07/ 6.4e+05
find max Bottom	F	1.7e+08/ 9.5e+07	2.5e+08 /4.1e+08	2.4e+06/ 1.0e+06	2.9e+02/ 7.3e+01	1.5e+05/ 1.1e+04	3.0e+02/ 7.6e+01	1.4e+07/ 1.4e+06
scaleLayout	M	5.9e+07 /9.0e+07	8.8e+07 /2.9e+08	1.1e+06/ 7.5e+05	5.8e+04/ 4.5e+04	1.5e+04/ 8.7e+03	1.4e+03 /1.4e+04	1.3e+06/ 4.5e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 1.9e+04	5.5e+02/ 7.6e+01	4.2e+05/ 4.2e+03	5.5e+02/ 7.5e+01	1.4e+07/ 6.4e+05
Total		8.1e+08/5.4e+08	1.2e+09/2.2e+09	1.2e+07/3.8e+06	1.1e+05/1.1e+05	1.1e+06/5.1e+04	4.3e+03/3.3e+04	5.7e+07/5.3e+06

Table 17: Per-pass PAPI counters for DomTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.5e+08 /4.6e+08	4.4e+08 /1.8e+09	8.2e+06/ 2.2e+06	2.0e+02 /8.8e+02	1.2e+04 /3.2e+04	2.0e+02 /8.8e+02	1.0e+07/ 1.3e+06
computeWidths	M	6.7e+07 /3.5e+08	1.5e+08 /9.0e+08	1.4e+06 /3.2e+06	4.6e+04 /2.4e+05	5.9e+03 /1.6e+04	1.1e+03 /2.3e+04	1.2e+06/ 4.3e+05
count styled	F	3.5e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.2e+06	1.3e+02 /3.6e+02	3.5e+04 /3.7e+04	1.3e+02 /3.2e+02	1.0e+07/ 4.1e+05
find max Bottom	F	3.4e+08 /4.5e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.6e+06	8.0e+01 /6.9e+02	1.4e+04 /2.6e+04	8.0e+01 /6.9e+02	1.0e+07/ 9.5e+05
scaleLayout	M	6.7e+07 /3.5e+08	1.1e+08 /7.6e+08	9.8e+05 /2.8e+06	5.1e+04 /8.5e+04	3.8e+03 /1.2e+04	1.2e+03 /2.4e+04	1.1e+06/ 4.6e+05
sumTextWidth	F	3.5e+08 /4.5e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	3.9e+02 /8.4e+02	8.7e+04/ 3.9e+04	3.9e+02 /6.9e+02	1.0e+07/ 4.1e+05
Total		1.5e+09/2.5e+09	2.0e+09/8.6e+09	4.1e+07/1.2e+07	9.8e+04/3.3e+05	1.6e+05/1.6e+05	3.1e+03/5.0e+04	4.3e+07/4.0e+06

Table 18: Per-pass performance for KDTree, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0270 $\pm$ 0.00004	0.0390 $\pm$ 0.00005	0.0210$\pm$0.00003	0.1178 $\pm$ 0.00129	1.28$\times$	1.45$\times$	1.86$\times$	0.33 $\times$
countInRange tight_box	F	13/17	24%	0.0272 $\pm$ 0.00001	0.0402 $\pm$ 0.00001	0.0195$\pm$0.00001	0.1236 $\pm$ 0.00131	1.39$\times$	1.48$\times$	2.06$\times$	0.32 $\times$
photonMappingBenchmark	F	12/17	29%	0.0417 $\pm$ 0.00108	0.0473 $\pm$ 0.00001	0.0409$\pm$0.00001	0.1303 $\pm$ 0.00164	1.02 $\times$	1.13$\times$	1.16$\times$	0.36 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0241 $\pm$ 0.00001	0.0380 $\pm$ 0.00014	0.0162$\pm$0.00000	0.1142 $\pm$ 0.00003	1.49$\times$	1.58$\times$	2.34$\times$	0.33 $\times$
sumMassInRange	F	14/17	18%	0.0265 $\pm$ 0.00002	0.0402 $\pm$ 0.00001	0.0199$\pm$0.00000	0.1233 $\pm$ 0.00001	1.33$\times$	1.52$\times$	2.02$\times$	0.33 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0281 $\pm$ 0.00001	0.0388 $\pm$ 0.00108	0.0208$\pm$0.00000	0.1135 $\pm$ 0.00020	1.35$\times$	1.38$\times$	1.87$\times$	0.34 $\times$
Total				0.1745	0.2435	0.1383	0.7226	1.26$\times$	1.40$\times$	1.76$\times$	0.34 $\times$
Geomean								1.30$\times$	1.41$\times$	1.84$\times$	0.34 $\times$

Table 19: Per-pass PAPI counters for KDTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.9e+06/ 1.6e+06	1.7e+02/ 1.4e+02	4.4e+03 /1.7e+04	1.7e+02/ 1.5e+02	7.7e+06/ 1.6e+06
countInRange tight_box	F	1.4e+08/ 9.9e+07	4.0e+08 /5.5e+08	2.2e+06/ 1.8e+06	1.4e+02/ 1.0e+02	3.4e+03 /1.6e+04	1.4e+02/ 1.0e+02	7.7e+06/ 1.6e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	2.1e+02 /4.1e+02	3.3e+03 /1.4e+04	2.1e+02 /4.1e+02	5.6e+06/ 1.4e+06
pointCloudNeighborhood	F	1.2e+08/ 8.2e+07	3.3e+08 /4.8e+08	2.6e+06/ 2.0e+06	8.2e+01 /9.0e+01	3.9e+03 /1.5e+04	8.2e+01 /9.0e+01	7.8e+06/ 1.3e+06
sumMassInRange	F	1.3e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.6e+06 /3.0e+06	8.1e+01 /1.2e+02	3.7e+03 /1.8e+04	8.3e+01 /1.2e+02	7.7e+06/ 1.7e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.1e+06/ 6.1e+05	9.6e+01 /1.1e+02	2.8e+03 /1.4e+04	9.6e+01 /1.1e+02	7.7e+06/ 1.2e+06
Total		8.8e+08/7.0e+08	3.1e+09/4.0e+09	1.5e+07/1.2e+07	7.7e+02/9.8e+02	2.1e+04/9.5e+04	7.8e+02/9.8e+02	4.4e+07/8.9e+06

Table 20: Per-pass PAPI counters for KDTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.0e+08	4.9e+08 /1.3e+09	4.7e+06 /5.3e+06	1.1e+02 /1.4e+03	4.5e+03 /1.9e+04	1.1e+02 /1.4e+03	5.8e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.3e+08	4.5e+08 /1.3e+09	5.2e+06 /5.9e+06	1.5e+02 /1.9e+03	3.5e+04/ 1.9e+04	1.5e+02 /1.9e+03	5.6e+06/ 1.5e+06
photonMappingBenchmark	F	2.4e+08 /6.6e+08	1.1e+09 /2.0e+09	2.8e+06 /5.9e+06	3.9e+02 /1.5e+03	2.7e+03 /2.3e+04	3.9e+02 /1.5e+03	5.1e+06/ 1.5e+06
pointCloudNeighborhood	F	1.9e+08 /5.8e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.8e+06	2.0e+02 /1.2e+03	5.3e+03 /1.7e+04	2.0e+02 /1.2e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.3e+08	4.7e+08 /1.3e+09	5.1e+06 /6.4e+06	1.6e+02 /1.5e+03	3.3e+04/ 2.0e+04	1.6e+02 /1.5e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.8e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	2.5e+02 /1.3e+03	7.8e+03 /1.6e+04	2.5e+02 /1.3e+03	5.8e+06/ 1.3e+06
Total		1.2e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	1.3e+03/8.9e+03	8.9e+04/1.1e+05	1.3e+03/8.9e+03	3.4e+07/8.7e+06

Table 21: Per-pass performance for LinearListReduction, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0289 $\pm$ 0.00002	0.1391 $\pm$ 0.00174	0.0049$\pm$0.00001	<i>OOM</i>	5.84$\times$	4.82$\times$	28.11$\times$	–
Total				0.0289	0.1391	0.0049	–	5.84$\times$	4.82$\times$	28.11$\times$	–
Geomean								5.84$\times$	4.82$\times$	28.11$\times$	–

Table 22: Per-pass PAPI counters for LinearListReduction (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	2.3e+02/ 4.3e+01	2.7e+05/ 6.9e+03	2.3e+02/ 4.4e+01	1.2e+07/ 1.1e+06
Total		1.5e+08/2.5e+07	9.0e+07/1.0e+08	1.5e+06/4.6e+04	2.3e+02/4.3e+01	2.7e+05/6.9e+03	2.3e+02/4.4e+01	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for LinearListReduction (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	7.1e+08/–	2.5e+08/–	5.0e+06/–	4.4e+02/–	5.3e+05/–	4.5e+02/–	2.6e+07/–
Total		7.1e+08/–	2.5e+08/–	5.0e+06/–	4.4e+02/–	5.3e+05/–	4.5e+02/–	2.6e+07/–

Table 24: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2508 $\pm$ 0.00204	<i>OOM</i>	0.3392 $\pm$ 0.00213	<i>OOM</i>	0.74 $\times$	–	–	–
length List	M	1/2	50%	0.0413 $\pm$ 0.00001	<i>OOM</i>	0.0215 $\pm$ 0.00003	<i>OOM</i>	1.92 $\times$	–	–	–
sumList	F	2/2	0%	0.0441 $\pm$ 0.00002	<i>OOM</i>	0.0451 $\pm$ 0.00022	<i>OOM</i>	0.98 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0443 $\pm$ 0.00002	<i>OOM</i>	0.0445 $\pm$ 0.00001	<i>OOM</i>	1.00 $\times$	–	–	–
Total				0.3805	–	0.4503	–	0.85 $\times$	–	–	–
Geomean								1.08 $\times$	–	–	–

Table 25: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.6e+08	1.9e+09 /2.8e+09	8.0e+06/ 1.4e+06	3.0e+05 /5.9e+05	1.1e+04 /5.6e+04	9.2e+03 /1.4e+04	9.1e+06/ 7.8e+06
length List	M	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.1e+05/ 8.8e+04	1.0e+02/ 7.9e+01	2.7e+03 /1.4e+04	1.0e+02/ 8.0e+01	1.3e+07/ 1.1e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	6.8e+05/ 5.4e+05	8.9e+01/ 7.9e+01	3.1e+03 /6.8e+04	9.0e+01/ 8.0e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.5e+05 /5.6e+05	1.3e+02 /1.4e+02	2.8e+03 /5.8e+04	1.4e+02 /1.4e+02	1.3e+07/ 1.2e+07
Total		1.2e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.6e+06	3.0e+05/5.9e+05	1.9e+04/2.0e+05	9.6e+03/1.4e+04	4.8e+07/3.3e+07

Table 26: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0453 $\pm$ 0.00040	0.0604 $\pm$ 0.00041	0.0428 $\pm$ 0.00043	0.1125 $\pm$ 0.00117	1.06 $\times$	1.33 $\times$	1.41 $\times$	0.54 $\times$
sumTree	F	3/3	0%	0.0090 $\pm$ 0.00000	0.0401 $\pm$ 0.00001	0.0113 $\pm$ 0.00001	0.0805 $\pm$ 0.00002	0.79 $\times$	4.48 $\times$	3.55 $\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0083 $\pm$ 0.00000	0.0401 $\pm$ 0.00115	0.0111 $\pm$ 0.00002	0.0806 $\pm$ 0.00002	0.75 $\times$	4.85 $\times$	3.63 $\times$	0.50 $\times$
Total				0.0626	0.1407	0.0652	0.2735	0.96 $\times$	2.25 $\times$	2.16 $\times$	0.51 $\times$
Geomean								0.86 $\times$	3.07 $\times$	2.63 $\times$	0.51 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.2e+05 /4.0e+05	3.5e+04 /4.5e+04	1.9e+03 /5.0e+03	1.2e+03 /1.7e+03	5.1e+05/ 4.5e+05
sumTree	F	4.6e+07 /5.7e+07	2.2e+08 /2.6e+08	1.0e+04 /3.3e+04	1.5e+02 /2.5e+02	1.1e+03 /3.9e+03	1.5e+02 /2.5e+02	7.6e+05/ 6.1e+05
sumTree TailRec	F	4.2e+07 /5.6e+07	1.9e+08 /2.2e+08	5.8e+03 /1.9e+04	1.6e+02 /3.1e+02	6.8e+02 /3.9e+03	1.7e+02 /3.1e+02	7.7e+05/ 6.0e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.4e+05/4.6e+05	3.5e+04/4.6e+04	3.7e+03/1.3e+04	1.5e+03/2.3e+03	2.0e+06/1.7e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	1.4e+05 /9.7e+05	3.2e+04 /5.0e+04	1.7e+03 /1.6e+04	1.3e+03 /3.0e+03	4.1e+05/ 4.0e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	7.6e+04 /8.9e+05	1.8e+02 /3.1e+02	1.0e+03 /1.3e+04	1.8e+02 /2.5e+02	3.3e+05 /3.5e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.4e+04 /8.7e+05	5.1e+02/ 9.2e+01	1.1e+03 /1.3e+04	5.0e+02/ 8.9e+01	3.3e+05 /3.5e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	2.8e+05/2.7e+06	3.3e+04/5.1e+04	3.8e+03/4.1e+04	2.0e+03/3.4e+03	1.1e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0122 $\pm$ 0.00000	0.0444 $\pm$ 0.00001	0.0103 $\pm$ 0.00002	0.0464 $\pm$ 0.00114	1.18 $\times$	3.62 $\times$	4.29 $\times$	0.96 $\times$
countMarked	F	3/5	40%	0.0123 $\pm$ 0.00000	0.0445 $\pm$ 0.00001	0.0108 $\pm$ 0.00002	0.0467 $\pm$ 0.00002	1.13 $\times$	3.64 $\times$	4.12 $\times$	0.95 $\times$
countSurvivors	F	4/5	20%	0.0133 $\pm$ 0.00068	0.0447 $\pm$ 0.00173	0.0131 $\pm$ 0.00001	0.0468 $\pm$ 0.00001	1.01 $\times$	3.37 $\times$	3.42 $\times$	0.96 $\times$
deadBytes	F	4/5	20%	0.0129 $\pm$ 0.00003	0.0440 $\pm$ 0.00002	0.0125 $\pm$ 0.00001	0.0470 $\pm$ 0.00001	1.03 $\times$	3.40 $\times$	3.51 $\times$	0.94 $\times$
liveBytes	F	4/5	20%	0.0129 $\pm$ 0.00068	0.0433 $\pm$ 0.00132	0.0126 $\pm$ 0.00000	0.0470 $\pm$ 0.00001	1.02 $\times$	3.36 $\times$	3.43 $\times$	0.92 $\times$
sumObjIds	F	3/5	40%	0.0122 $\pm$ 0.00001	0.0432 $\pm$ 0.00001	0.0104 $\pm$ 0.00002	0.0471 $\pm$ 0.00015	1.17 $\times$	3.55 $\times$	4.14 $\times$	0.92 $\times$
sweepUnmarked	M	5/5	0%	0.0718 $\pm$ 0.00006	0.0871 $\pm$ 0.00006	0.0807 $\pm$ 0.00114	0.1858 $\pm$ 0.00007	0.89 $\times$	1.21 $\times$	1.08 $\times$	0.47 $\times$
totalHeapSize	F	3/5	40%	0.0120 $\pm$ 0.00011	0.0432 $\pm$ 0.00127	0.0106 $\pm$ 0.00010	0.0472 $\pm$ 0.00004	1.14 $\times$	3.58 $\times$	4.07 $\times$	0.92 $\times$
touchHotObjects	M	5/5	0%	0.0707 $\pm$ 0.00108	0.0868 $\pm$ 0.00022	0.0804 $\pm$ 0.00002	0.1898 $\pm$ 0.00247	0.88 $\times$	1.23 $\times$	1.08 $\times$	0.46 $\times$
Total				0.2303	0.4812	0.2415	0.7037	0.95 $\times$	2.09 $\times$	1.99 $\times$	0.68 $\times$
Geomean								1.05 $\times$	2.77 $\times$	2.90 $\times$	0.80 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.2e+07/ 5.2e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.4e+04	3.7e+02/ 1.4e+02	3.3e+03 /3.7e+03	3.7e+02/ 1.4e+02	2.9e+06/ 6.5e+05
countMarked	F	6.2e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.2e+04	2.6e+02/ 1.7e+02	3.2e+03 /3.9e+03	2.6e+02/ 1.7e+02	2.9e+06/ 6.4e+05
countSurvivors	F	6.7e+07/ 6.6e+07	3.1e+08 /3.8e+08	9.9e+05/ 6.9e+04	3.3e+02/ 1.8e+02	2.4e+03 /5.3e+03	3.3e+02/ 1.8e+02	2.7e+06/ 1.2e+06
deadBytes	F	6.6e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.2e+05/ 5.8e+04	2.1e+02/ 1.5e+02	2.3e+03 /5.3e+03	2.1e+02/ 1.5e+02	2.7e+06/ 1.2e+06
liveBytes	F	6.5e+07/ 6.4e+07	2.7e+08 /3.5e+08	8.8e+05/ 5.8e+04	2.1e+02/ 1.7e+02	2.4e+03 /5.4e+03	2.1e+02/ 1.7e+02	2.8e+06/ 1.2e+06
sumObjIds	F	6.2e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.1e+06/ 7.6e+04	3.2e+02/ 1.1e+02	2.8e+03 /4.3e+03	3.2e+02/ 1.1e+02	2.8e+06/ 6.2e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	9.8e+05/ 8.0e+05	1.1e+05/ 1.1e+05	5.3e+03 /1.4e+04	5.0e+03 /6.9e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.1e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.2e+06/ 8.6e+04	1.7e+02/ 1.3e+02	2.4e+03 /3.6e+03	1.7e+02/ 1.3e+02	2.9e+06/ 6.4e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	8.1e+05/ 5.5e+05	1.0e+05 /1.5e+05	5.8e+03 /1.1e+04	4.5e+03 /1.0e+04	2.0e+06/ 1.1e+06
Total		8.2e+08/8.8e+08	3.0e+09/4.0e+09	9.0e+06/1.8e+06	2.1e+05/2.6e+05	3.0e+04/5.6e+04	1.1e+04/1.8e+04	2.4e+07/8.3e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.8e+05	2.4e+02/ 6.2e+01	4.9e+03 /7.9e+03	2.4e+02/ 6.3e+01	1.7e+06/ 3.7e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.9e+05	1.2e+02/ 8.7e+01	5.1e+03 /9.6e+03	1.2e+02/ 8.5e+01	1.7e+06/ 3.8e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /4.2e+05	1.6e+02 /2.6e+02	1.8e+03 /6.5e+03	1.6e+02 /2.6e+02	1.7e+06/ 6.9e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.8e+05 /4.4e+05	1.1e+02 /3.2e+02	3.8e+03 /6.4e+03	1.1e+02 /3.2e+02	1.7e+06/ 6.7e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.6e+05 /4.3e+05	1.3e+02 /3.0e+02	2.1e+03 /6.3e+03	1.3e+02 /3.0e+02	1.8e+06/ 6.8e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 5.0e+05	2.2e+02/ 8.3e+01	4.9e+03 /8.9e+03	2.3e+02/ 8.3e+01	1.7e+06/ 3.6e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	1.7e+05 /2.4e+06	7.8e+04 /1.4e+05	3.8e+03 /2.2e+04	2.2e+03 /5.1e+03	1.8e+06/ 9.2e+05
totalHeapSize	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.4e+05 /5.5e+05	9.1e+01 /3.5e+02	4.8e+03 /9.2e+03	9.2e+01 /3.5e+02	1.8e+06/ 3.8e+05
touchHotObjects	M	2.7e+08 /7.9e+08	7.7e+08 /2.2e+09	1.8e+05 /2.0e+06	8.9e+04 /1.3e+05	3.8e+03 /2.1e+04	2.7e+03 /6.2e+03	1.8e+06/ 9.7e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.3e+06/7.7e+06	1.7e+05/2.7e+05	3.5e+04/9.7e+04	6.0e+03/1.3e+04	1.6e+07/5.4e+06

Table 33: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1195$\pm$0.00109	0.1300 $\pm$ 0.00011	0.1333 $\pm$ 0.00156	0.4056 $\pm$ 0.00299	0.90 $\times$	1.09 $\times$	0.97 $\times$	0.32 $\times$
autorefineMaxLevel	F	4/8	50%	0.0203 $\pm$ 0.00000	0.0448 $\pm$ 0.00001	0.0194$\pm$0.00085	0.2347 $\pm$ 0.00174	1.04 $\times$	2.21$\times$	2.31$\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0194 $\pm$ 0.00060	0.0453 $\pm$ 0.00001	0.0114$\pm$0.00001	0.2338 $\pm$ 0.00089	1.70$\times$	2.33$\times$	3.96$\times$	0.19 $\times$
diffPW	M	8/8	0%	0.1201$\pm$0.00013	0.1332 $\pm$ 0.00147	0.1308 $\pm$ 0.00113	0.4073 $\pm$ 0.00135	0.92 $\times$	1.11$\times$	1.02 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0211 $\pm$ 0.00001	0.0470 $\pm$ 0.00002	0.0184$\pm$0.00001	0.2368 $\pm$ 0.00164	1.15$\times$	2.22$\times$	2.56$\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0247 $\pm$ 0.00003	0.0544 $\pm$ 0.00008	0.0175$\pm$0.00005	0.1980 $\pm$ 0.00128	1.41$\times$	2.21$\times$	3.11$\times$	0.27 $\times$
pmapCutHistogram	F	4/8	50%	0.0316 $\pm$ 0.00001	0.0552 $\pm$ 0.00114	0.0123$\pm$0.00001	0.2421 $\pm$ 0.00284	2.56$\times$	1.75$\times$	4.48$\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0197 $\pm$ 0.00001	0.0459 $\pm$ 0.00001	0.0114$\pm$0.00001	0.2022 $\pm$ 0.00131	1.72$\times$	2.34$\times$	4.02$\times$	0.23 $\times$
Total				0.3764	0.5558	0.3546	2.161	1.06 $\times$	1.48$\times$	1.57$\times$	0.26 $\times$
Geomean								1.34$\times$	1.83$\times$	2.45$\times$	0.24 $\times$

Table 34: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.4e+08	5.7e+08 /1.2e+09	1.4e+06 /2.1e+06	2.9e+05/ 2.7e+05	1.0e+04 /2.2e+04	8.2e+03 /2.2e+04	4.4e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.8e+07	2.5e+08 /3.7e+08	2.5e+06/ 1.9e+05	3.8e+02/ 4.9e+01	6.5e+03/ 6.2e+03	3.8e+02/ 4.8e+01	6.1e+06/ 9.0e+05
compressMass	F	9.9e+07/ 5.8e+07	2.5e+08 /2.8e+08	2.1e+06/ 3.0e+04	7.4e+02/ 2.8e+02	2.0e+04/ 4.8e+03	7.4e+02/ 2.8e+02	6.2e+06/ 6.1e+05
diffPW	M	2.8e+08 /3.4e+08	5.9e+08 /1.3e+09	1.1e+06 /2.1e+06	1.6e+05 /2.0e+05	1.1e+04 /2.5e+04	9.8e+03 /2.3e+04	4.4e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.3e+07	3.3e+08 /4.9e+08	2.5e+06/ 2.9e+05	4.2e+02/ 1.7e+02	1.5e+04/ 8.7e+03	4.2e+02/ 1.7e+02	6.2e+06/ 1.5e+06
norm2Estimate	F	1.2e+08/ 8.9e+07	3.4e+08 /4.4e+08	1.4e+06/ 3.8e+05	2.4e+02/ 1.2e+02	1.1e+04/ 9.2e+03	2.5e+02/ 1.2e+02	6.1e+06/ 1.5e+06
pmapCutHistogram	F	1.6e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.3e+04	1.6e+03/ 4.1e+02	1.0e+05/ 7.3e+03	1.6e+03/ 4.1e+02	4.7e+06/ 1.2e+06
truncateTolViolations	F	1.0e+08/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 3.2e+04	6.3e+02/ 3.6e+02	1.8e+04/ 4.8e+03	6.3e+02/ 3.6e+02	6.3e+06/ 6.2e+05
Total		1.2e+09/1.1e+09	2.8e+09/4.7e+09	1.5e+07/5.2e+06	4.5e+05/4.8e+05	2.0e+05/8.8e+04	2.2e+04/4.6e+04	4.4e+07/9.9e+06

Table 35: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	2.9e+05 /5.1e+06	1.3e+05 /3.6e+05	5.3e+03 /3.8e+04	4.3e+03 /3.3e+04	4.2e+06/ 2.0e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	2.3e+02 /1.3e+03	1.5e+03 /4.1e+04	2.3e+02 /4.9e+02	4.1e+06/ 6.8e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	1.8e+02 /1.5e+03	1.5e+03 /3.0e+04	1.8e+02 /2.4e+02	4.1e+06/ 3.7e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	2.9e+05 /5.1e+06	1.2e+05 /2.7e+05	5.6e+03 /3.6e+04	4.0e+03 /2.1e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.5e+05 /2.6e+06	1.4e+02 /3.1e+02	1.6e+03 /4.3e+04	1.4e+02 /3.0e+02	4.1e+06/ 9.9e+05
norm2Estimate	F	2.8e+08 /1.0e+09	5.1e+08 /1.3e+09	2.5e+05 /2.5e+06	9.7e+01 /7.0e+02	1.4e+03 /4.3e+04	9.7e+01 /5.8e+02	3.9e+06/ 9.6e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.5e+05 /2.0e+06	9.6e+02 /2.1e+03	1.0e+04 /3.8e+04	9.7e+02/ 5.9e+02	3.9e+06/ 6.7e+05
truncateTolViolations	F	2.3e+08 /1.0e+09	4.4e+08 /1.3e+09	3.2e+05 /1.3e+06	3.9e+02 /3.8e+03	3.6e+03 /2.7e+04	3.9e+02 /1.0e+03	4.0e+06/ 3.7e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	2.6e+06/2.2e+07	2.5e+05/6.4e+05	3.1e+04/2.9e+05	1.0e+04/5.8e+04	3.2e+07/7.8e+06

Table 36: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0829 $\pm$ 0.00015	0.0933 $\pm$ 0.00021	0.0800$\pm$0.00289	0.3410 $\pm$ 0.00173	1.04 $\times$	1.12$\times$	1.17$\times$	0.27 $\times$
sum tree	F	5/5	0%	0.0278 $\pm$ 0.00099	0.0503 $\pm$ 0.00002	0.0256$\pm$0.00008	0.2768 $\pm$ 0.00211	1.09 $\times$	1.81$\times$	1.96$\times$	0.18 $\times$
Total				0.1108	0.1436	0.1056	0.6178	1.05 $\times$	1.30$\times$	1.36$\times$	0.23 $\times$
Geomean								1.06 $\times$	1.43$\times$	1.51$\times$	0.22 $\times$

Table 37: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.3e+05 /1.2e+06	6.3e+04 /1.7e+05	3.4e+03 /1.2e+04	2.5e+03/ 9.0e+02	1.5e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	7.9e+04 /4.1e+05	6.0e+01 /6.9e+01	7.8e+02 /9.1e+03	6.1e+01 /6.9e+01	1.7e+06/ 1.3e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	5.1e+05/1.6e+06	6.3e+04/1.7e+05	4.2e+03/2.1e+04	2.6e+03/9.6e+02	3.3e+06/2.4e+06

Table 38: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	2.0e+05 /2.3e+06	6.7e+04 /1.9e+05	2.9e+03 /4.4e+04	3.2e+03 /5.5e+03	1.5e+06/ 9.6e+05
sum tree	F	2.6e+08 /1.4e+09	5.8e+08 /1.0e+09	8.8e+04 /2.4e+06	9.7e+01 /1.1e+03	1.4e+03 /4.2e+04	1.0e+02 /2.6e+02	1.3e+06/ 8.7e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	2.9e+05/4.7e+06	6.7e+04/1.9e+05	4.3e+03/8.6e+04	3.3e+03/5.7e+03	2.8e+06/1.8e+06

Table 39: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0147 $\pm$ 0.00001	0.0272 $\pm$ 0.00000	0.0082$\pm$0.00003	0.1077 $\pm$ 0.00004	1.80$\times$	1.85$\times$	3.34$\times$	0.25 $\times$
countLazyNodes	F	4/10	60%	0.0126 $\pm$ 0.00000	0.0275 $\pm$ 0.00001	0.0066$\pm$0.00000	0.1107 $\pm$ 0.00117	1.91$\times$	2.19$\times$	4.19$\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0119 $\pm$ 0.00000	0.0256 $\pm$ 0.00015	0.0058$\pm$0.00000	0.1113 $\pm$ 0.00002	2.04$\times$	2.15$\times$	4.40$\times$	0.23 $\times$
decayTrieStats	M	10/10	0%	0.0742$\pm$0.00092	0.0801 $\pm$ 0.00122	0.0842 $\pm$ 0.00012	0.2393 $\pm$ 0.00120	0.88 $\times$	1.08 $\times$	0.95 $\times$	0.33 $\times$
resetTraversalState	M	10/10	0%	0.0752$\pm$0.00012	0.0815 $\pm$ 0.00105	0.0823 $\pm$ 0.00116	0.2387 $\pm$ 0.00144	0.91 $\times$	1.08 $\times$	0.99 $\times$	0.34 $\times$
sumPrefixFreq	F	3/10	70%	0.0120 $\pm$ 0.00007	0.0261 $\pm$ 0.00103	0.0057$\pm$0.00008	0.1212 $\pm$ 0.00170	2.09$\times$	2.18$\times$	4.54$\times$	0.22 $\times$
sumSubtreeHints	F	3/10	70%	0.0120 $\pm$ 0.00001	0.0260 $\pm$ 0.00000	0.0057$\pm$0.00000	0.1212 $\pm$ 0.00018	2.09$\times$	2.17$\times$	4.54$\times$	0.21 $\times$
Total				0.2125	0.2939	0.1985	1.050	1.07 $\times$	1.38$\times$	1.48$\times$	0.28 $\times$
Geomean								1.58$\times$	1.74$\times$	2.75$\times$	0.26 $\times$

Table 40: Per-pass PAPI counters for **Trie** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.4e+07/ 4.1e+07	1.3e+08 /2.1e+08	1.5e+06/ 4.8e+05	6.0e+01/ 4.4e+01	5.4e+04/ 4.7e+03	6.7e+01/ 4.5e+01	3.6e+06/ 7.6e+05
countLazyNodes	F	6.4e+07/ 3.3e+07	1.3e+08 /1.6e+08	1.4e+06/ 3.1e+04	1.0e+02/ 8.6e+01	1.6e+04/ 3.3e+03	1.1e+02/ 9.0e+01	4.0e+06/ 5.6e+05
countTerminals	F	6.0e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.5e+06/ 1.1e+04	1.3e+02/ 6.2e+01	2.5e+04/ 2.3e+03	1.3e+02/ 6.6e+01	3.9e+06/ 2.2e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	9.4e+05 /1.4e+06	1.1e+05 /1.1e+05	6.0e+03 /2.1e+04	3.4e+03 /2.8e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.7e+08 /2.0e+08	2.8e+08 /7.0e+08	9.3e+05/ 7.7e+05	9.4e+04 /1.2e+05	6.8e+03 /1.5e+04	2.0e+03 /2.1e+04	3.1e+06/ 9.1e+05
sumPrefixFreq	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.3e+04	9.3e+01/ 6.1e+01	2.9e+04/ 2.5e+03	9.4e+01/ 6.4e+01	3.9e+06/ 2.0e+05
sumSubtreeHints	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.5e+04	1.3e+02/ 7.0e+01	2.7e+04/ 2.5e+03	1.4e+02/ 7.1e+01	3.9e+06/ 2.1e+05
Total		6.5e+08/5.7e+08	1.2e+09/2.2e+09	9.2e+06/2.7e+06	2.0e+05/2.3e+05	1.6e+05/5.2e+04	6.0e+03/5.0e+04	2.5e+07/4.1e+06

Table 41: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	1.2e+02 /7.0e+02	3.1e+03 /2.7e+04	1.2e+02 /5.8e+02	2.7e+06/ 4.6e+05
countLazyNodes	F	1.4e+08 /5.6e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	6.7e+01 /6.9e+02	2.7e+03 /2.2e+04	6.7e+01 /2.2e+02	2.6e+06/ 3.1e+05
countTerminals	F	1.3e+08 /5.7e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.4e+05	5.8e+01 /1.1e+03	1.7e+03 /1.6e+04	6.0e+01 /1.3e+02	2.8e+06/ 1.5e+05
decayTrieStats	M	1.9e+08 /1.0e+09	4.5e+08 /2.1e+09	3.0e+05 /5.2e+06	9.5e+04 /3.9e+05	3.4e+03 /3.4e+04	5.4e+03 /3.7e+04	2.8e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.9e+08	3.7e+08 /2.0e+09	3.3e+05 /3.3e+06	1.3e+05 /1.6e+05	3.9e+03 /2.7e+04	4.9e+03 /3.0e+04	2.9e+06/ 9.8e+05
sumPrefixFreq	F	1.3e+08 /6.2e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	1.4e+02 /9.0e+02	1.4e+03 /1.6e+04	1.4e+02/ 1.2e+02	2.8e+06/ 1.5e+05
sumSubtreeHints	F	1.3e+08 /6.2e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	7.0e+01 /9.0e+02	1.4e+03 /1.6e+04	7.7e+01 /1.2e+02	2.7e+06/ 1.5e+05
Total		1.1e+09/4.9e+09	1.9e+09/7.8e+09	7.8e+06/1.3e+07	2.2e+05/5.5e+05	1.8e+04/1.6e+05	1.1e+04/6.8e+04	1.9e+07/3.4e+06